

Validation, Backtesting, Audit & Compliance Plan

GSPS Implementation Specification Pack • Prepared for Claude Code • August 28, 2026 • Draft implementation directives; requires securities/compliance counsel review before live personalized recommendations or execution.

Validation before live use

Indicators are not strategies. Every GSPS rule set must be tested as a versioned, deterministic system using only information that would have been available when the signal was generated.

- Write exact rules before evaluating performance.
- Separate development, validation, and out-of-sample datasets.
- Use walk-forward testing; prohibit repeated parameter selection using the same test sample.
- Model spread, commissions, slippage, gaps, partial fills, corporate actions, and order behavior.
- Report results separately by strategy, market regime, volatility regime, sector, instrument group, and holding period.
- Perform stress tests for earnings gaps, broad-market selloffs, volatility spikes, and degraded liquidity.
- Freeze a strategy version before shadow/live-paper tracking.

Required performance metrics

Metric	Reason
Sample size and date range	Avoids claims from small/selected datasets
Win rate and average/median win	Shows positive-result distribution
Average/median loss and maximum loss	Shows loss severity
Expectancy	$\text{win probability} \times \text{average win} - \text{loss probability} \times \text{average loss}$
Profit factor	Gross profit divided by gross loss
Maximum drawdown	Shows peak-to-trough strategy risk
Time in trade and exposure	Shows capital/time requirements
Slippage sensitivity	Shows dependence on ideal execution
Out-of-sample results	Primary evidence against overfitting

If performance is shown to users, distinguish actual from hypothetical/backtested results, identify assumptions and limitations, and avoid cherry-picked results.

Audit and explainability requirements

- Persist input market-data snapshot/version, signal timestamp, strategy version, rule values, component score, eligibility results, displayed language, user actions, and imported execution results.
- Every user-visible score must have an explanation trace showing which conditions passed, failed, or were unknown.
- Keep immutable event records for cooldown/lock state changes and any policy override attempt.
- Version feature flags, policy thresholds, universe definitions, and copy; link each plan to the versions used.
- Implement data lineage: source, retrieval time, adjustment status, and confidence/freshness.

Compliance and legal workstream

Before providing live personalized trade recommendations, position-size instructions tied to user account data, broker account connectivity, or execution functionality, obtain specialist U.S. securities/compliance counsel. A disclaimer alone is not a complete product-boundary solution.

- Define whether GSPS is education, publication/research, analytics, alerts, or investment advice; review facts, marketing, compensation, and personalization.
- Create compliance review for copy, alerts, claims, testimonials, ranking labels, and hypothetical results.
- Create privacy/security policies for connected accounts, PII, market data, logs, consent, retention, and vendor access.
- Create incident/rollback procedures for data errors, strategy defects, stale signals, and erroneous notifications.
- Use clear AI governance: no claims of predictive certainty; human-approved policy changes; monitoring and review.

Launch gates

Gate	Must be complete before next stage
Internal alpha	Deterministic signal engine, audit logs, unit/integration tests, data freshness guardrails
Closed beta	Paper trading, user-plan adherence measurement, support workflows, performance monitoring
Live plan support	Legal/compliance review, broker/account disclosures, execution-risk communication, incident process
Broker automation	Separate approval: broker-specific capabilities, user consents, order safeguards, kill switches, reconciliation